Polymeric Chemistry
Rheology

Universität Stuttgart

Author 1:	Laurens Viehoff
Author 2:	Felix Roth
E-Mail 1:	st183688@stud.uni-stuttgart.de
E-Mail 2:	st188157@stud.uni-stuttgart.de
Student-ID 1:	3676761
Student-ID 2:	3711192

Assistent:	Semi Kim

Contents

1 Theory **2**

1.1 Basic concepts 2

2 Results **3**

2.1 Rotational measurements 3

2.2 Time dependant measurement of the viscosity 4

2.3 Oszillational measurements to determine the viscoelastic properties 5

2.3.1 amplitude sweep 5

2.3.2 frequency sweep 6

1 Theory

In rheology, the flow behaviour of a substance under different forces is studied. This behaviour can be determined using rotational or oscillatory measurements.

1.1 Basic concepts

To measure rheological properties, the sample is placed between two plates, the upper one of which is capable of moving. When this plate is set in motion, the sample experiences a force and moves along with it, but exhibits a velocity gradient from the lower to the upper plate. A visual representation of this is shown in Figure 1.

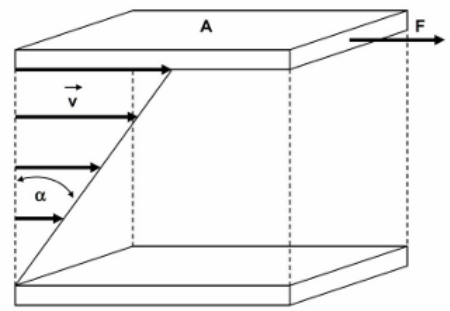

Figure 1: The diagram shows two plates with an area A, the upper one of which exerts a force F on the specimen, causing it to experience a velocity gradient $\vec{v}$ and to move through an angle α. [1]

Key parameters that can be detected or specified in this context include deformation γ, deformation rate $gam\dot{m}a$, shear stress τ and steady-state shear viscosity η. These parameters are defined by Equations (1–4).

$$\text{Deformation:} \qquad \gamma \equiv \frac{dx}{dy} = \tan\alpha \tag{1}$$

$$\text{Deformation rate:} \qquad \dot{\gamma} \equiv \frac{d\gamma}{dt} \tag{2}$$

$$\text{Shear stress:} \qquad \tau = \frac{F}{A} = \eta \cdot \dot{\gamma} \tag{3}$$

$$\text{Steady-state shear viscosity:} \qquad \eta = \frac{\tau}{\dot{\gamma}} \tag{4}$$

x and y represent the coordinate axes, with x pointing in the direction of the force F.
If a strain rate is specified and the shear stress is thereby detected, the viscosity can also be determined, allowing these parameters to be investigated as a function of the strain rate. Three types of curve behaviour can be detected for the shear stress or viscosity. The parameters may follow a linear trend or deviate from it. If the parameters deviate, they may become either larger or smaller at higher deformation rates. If a sample exhibits a linear relationship, it is referred to as a Newtonian fluid. If the parameters increase at higher deformation rates, it is referred to as a non-Newtonian shear-thickening fluid; conversely, if they decrease, it is referred to as a non-Newtonian shear-thinning fluid. The curves can be seen in Figure.

2 Results

2.1 Rotational measurements

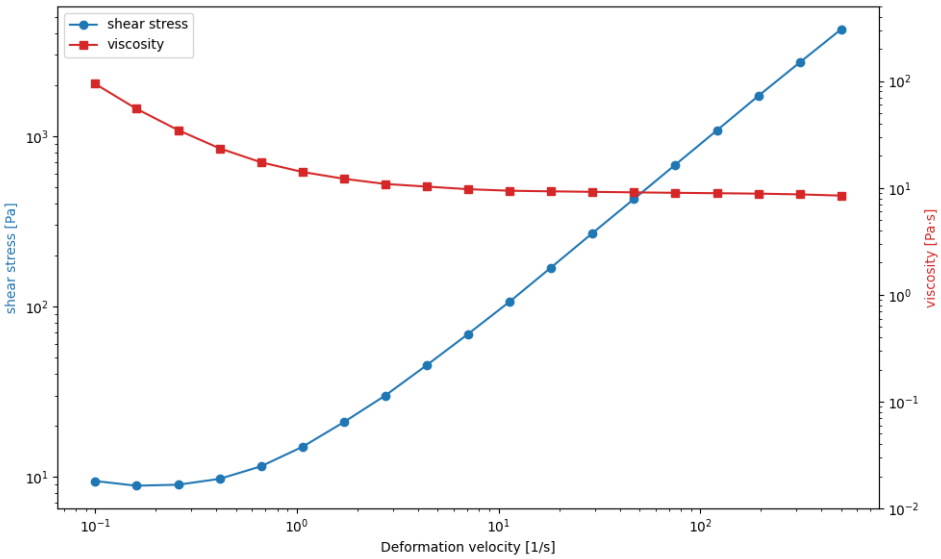

Figure 2: The graph shows the measurements of honey. On the left y-axis the shear stress and on the right y-axis the velocity is plotted, both of these are plotted against the deformation velocity.

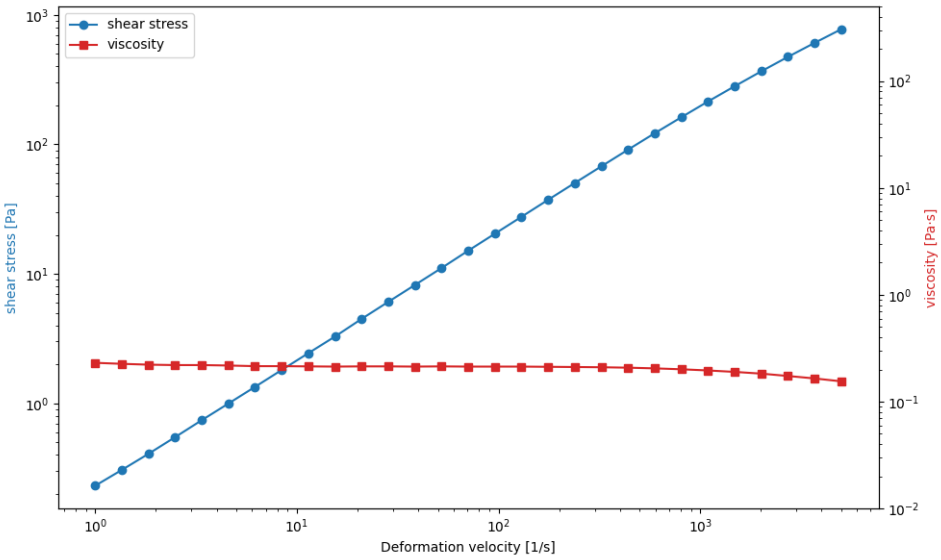

Figure 3: The graph shows the measurements of 14%-PVP solution. On the left y-axis the shear stress and on the right y-axis the velocity is plotted, both of these are plotted against the deformation velocity.

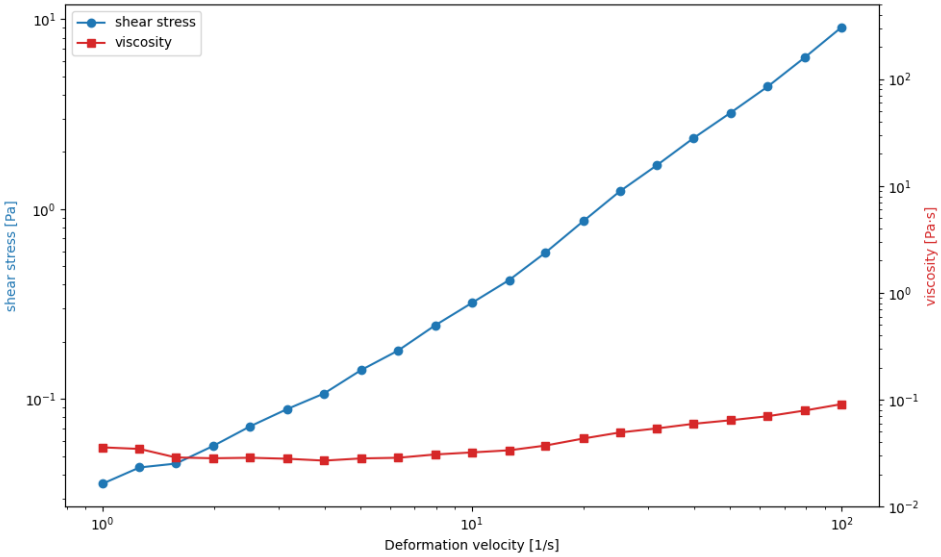

Figure 4: The graph shows the measurements of a starch solution. On the left y-axis the shear stress and on the right y-axis the velocity is plotted, both of these are plotted against the deformation velocity.

2.2 Time dependant measurement of the viscosity

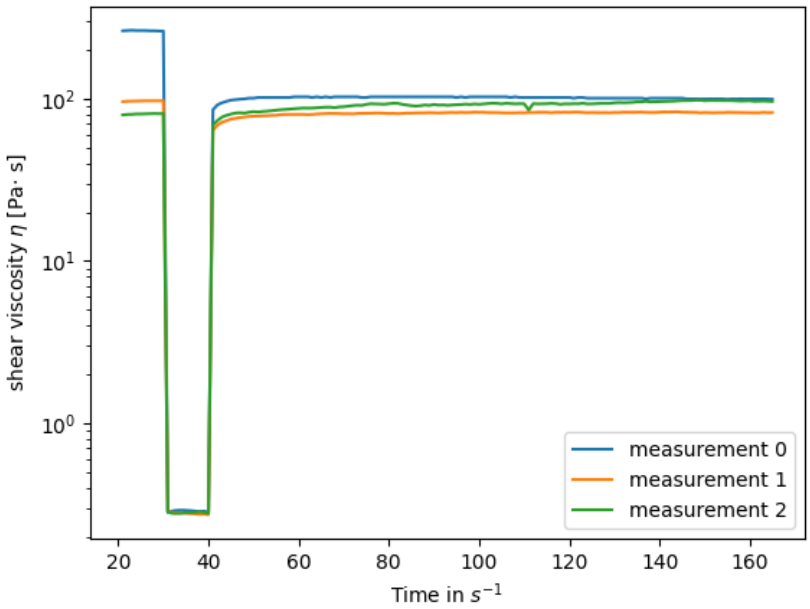

Figure 5: The graph shows the three time dependant viscosity measurements of ketchup. The shear viscosity is plotted against the duration.

2.3 Oszillational measurements to determine the viscoelastic properties

2.3.1 amplitude sweep

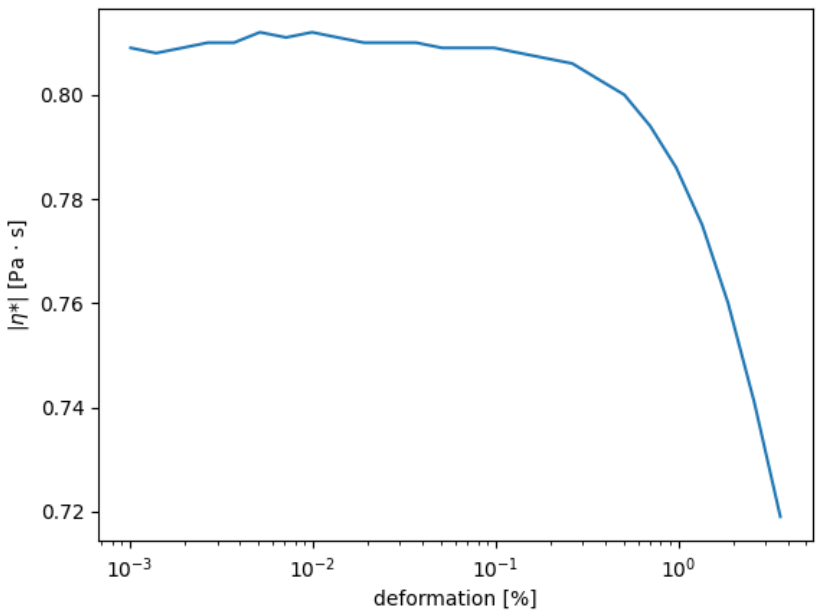

Figure 6: This graph shows the complex viscosity plotted against the deformation of 20%-PVP solution at constant frequency.

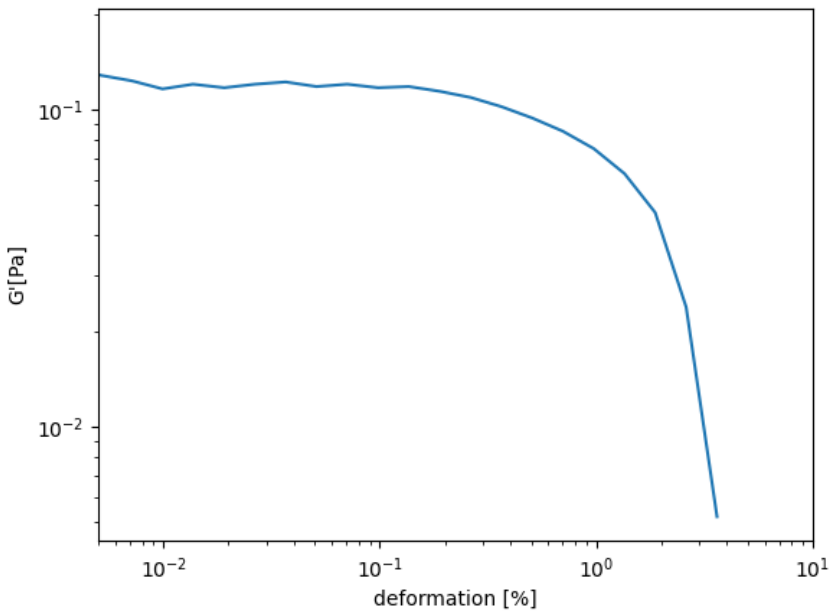

Figure 7: This graph shows the storage module G' plotted against the deformation of 20%-PVP solution at constant frequency.

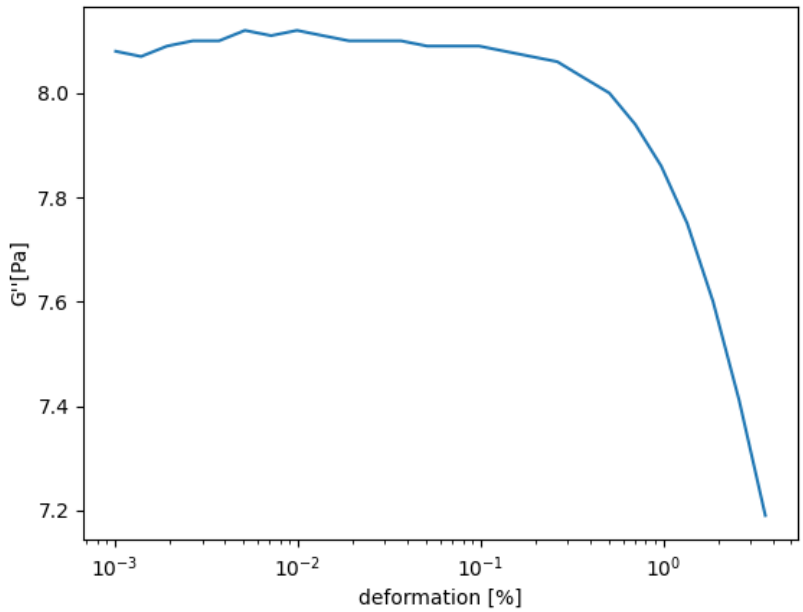

Figure 8: This graph shows the loss module G" plotted against the deformation of 20%-PVP solution at constant frequency.

2.3.2 frequency sweep

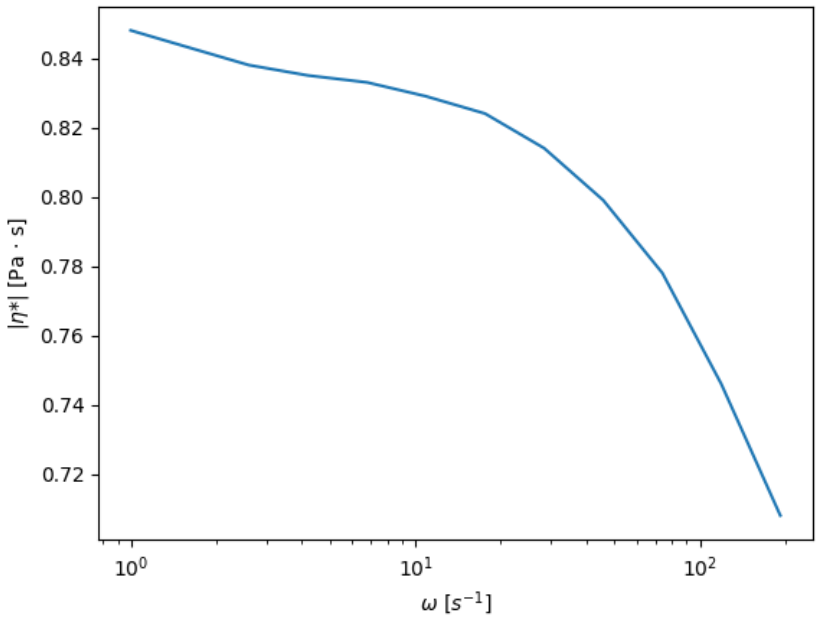

Figure 9: This graph shows the complex viscosity plotted against the deformation of 20%-PVP solution at constant amplitude.

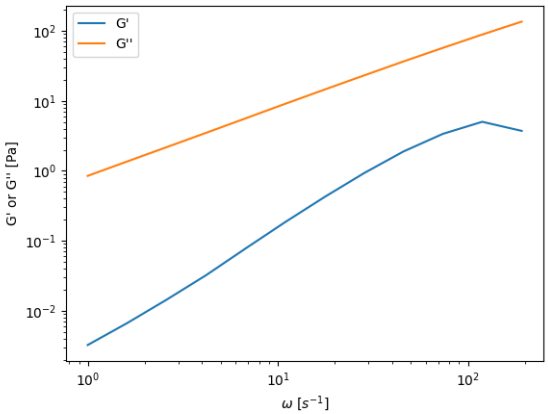

Figure 10: This graph shows the storage module G' loss module G" plotted against the deformation of 20%-PVP solution at constant amplitude.

References

[1] Institut für Polymerchemie - Polymerpraktikum, Vorlesungsskript, Universität Stuttgart, **März 2025**.